INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, NAGPUR
Department of Computer Science and Engineering
CSL 207: **Operating Systems**
Sessional 1 Exam
Semester –IV(CSE, DSA, AIML, HCIGT)

**Date:** 07/02/2024 (Wednesday)
**Time:** 10:00 to 11:00 AM

Duration: 1 Hour

Max. Marks: **15M**

*Important Instructions:*

- All the questions are compulsory.

Q1 Answer the following:

(a) On early computers, every byte of data read or written was handled by the CPU(there was no DMA). What implications does this have for multiprogramming?
(b) There are several services provided by the Operating system. Explain any two.
(c) Instructions related to IO devices are typically privileged, ie they are executed in kernel mode and not in user mode. Give reason why these are privileged.
(d) What are the differences between micro-kernel and monolithic kernel?
(e) What is the difference between kernel and user mode? Explain how having two distinct modes aid in designing an OS.

[5 M] [CO1, CO2]

Q2 Assume three processes: P1, P2 and P3 existing in a system. Times taken of execution of different processes are as follows: P1 = 10 ms, P2 = 7 ms and P3 = 15 ms. The processes are executing in a sequential manner such that P1 executes first, then P2, then P3, then again P1, P2 and P3, and so on, until a process completes its execution. Assume after 3ms the CPU switches from one process to another. Pictorially using process control block (PCB), explain how context switching is done.

[5 M] [CO2]

Q3 (a) Write a C program using file system calls that reads 10 characters from a text file and print on terminal. Skip next 5 characters and again read 10 characters and write on terminal.

[3 M] [CO3]

Q3 (b) Assume a process P1 is in running state and it encounters a printf() statement. What will be the next state transition of this process on execution of printf statement? Justify your answer with the help of state transition diagram.

[2 M] [CO2]

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